

SIMATS ENGINEERING

Department of Computer Science & Engineering

ASSESSMENT TOOL 1- INDUSTRY PROBLEM SOLVING TASK

Course Code:	CSA12	Course Title:	Computer Architecture for Multicore Systems
CO Assessed:	CO4	Assessment:	ASSESSMENT TOOL1- INDUSTRY PROBLEM SOLVING TASK
Weightage:	30%	Total Marks:	25
CO4 Statement:	<i>Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.</i>		
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ANNEXURE A

INDUSTRY PROBLEM SOLVING TASK

1. Smartphone Performance Optimization

A mobile manufacturer observes lag while switching between apps. Analyze the role of L1, L2, L3 cache and DRAM and propose a hierarchical memory redesign to reduce latency and improve throughput.

2. Cloud Server Memory Bottleneck

A cloud service provider experiences high latency in database queries. Compare cache hierarchy vs virtual memory paging and recommend improvements using appropriate memory technologies.

3. Autonomous Vehicle Data Processing
An autonomous vehicle must process sensor data in real time. Evaluate SRAM, DRAM, and MRAM and design a memory hierarchy that maximizes bandwidth and minimizes latency.
4. AI Inference Edge Device
An edge AI device needs fast model loading with low power consumption. Compare NAND Flash, DRAM, and RRAM and recommend an optimized hierarchical memory structure.
5. High-Performance Gaming Console
A gaming console suffers frame drops during large scene loading. Analyze L3 cache vs DRAM throughput and propose memory hierarchy enhancements.

Code of Conduct:

I D. Gnana Deepthi (192411123) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: D. Gnana Deepthi

ANSWERS

1. Smartphone Performance Optimization

Introduction

A mobile manufacturer observes noticeable lag when users switch between multiple applications. This lag occurs because the processor and applications require data and instructions from different levels of the memory hierarchy. When frequently required data is not available in fast cache memory, the processor must access slower levels of memory, eventually reaching DRAM. Therefore, an efficient hierarchical memory redesign is required to reduce latency and improve overall throughput.

Role of L1 Cache

The L1 cache is the smallest and fastest cache located closest to each CPU core. It stores the most frequently accessed instructions and data.

- Provides the lowest access latency.
- Usually divided into Instruction Cache (L1-I) and Data Cache (L1-D).
- Helps the processor execute instructions without frequently waiting for slower memory.
- A low L1 cache hit rate can cause processor stalls and affect app responsiveness.

Redesign: Increase L1 cache efficiency using better cache replacement policies, improved prefetching, and optimized cache size while maintaining very low access latency.

Role of L2 Cache

The L2 cache acts as the second level of cache when required data is not found in L1.

- Larger than L1 but slightly slower.
- Stores frequently reused data that cannot fit in L1.
- Reduces the number of accesses to the much slower L3 cache or DRAM.
- Can improve performance during application switching by retaining recently used application data.

Redesign: Use a larger private L2 cache for each performance core and optimize its replacement policy to retain recently accessed application data.

Role of L3 Cache

The **L3 cache** is generally larger and may be shared among multiple CPU cores.

- Allows different cores to share frequently used data.
- Reduces unnecessary accesses to DRAM.
- Helps when multiple applications and background processes run simultaneously.
- Improves inter-core communication and data sharing.

Redesign: Increase the shared L3 cache capacity and use intelligent cache partitioning or prioritization so that active applications receive sufficient cache space.

Role of DRAM

DRAM is the main memory of the smartphone and has much larger capacity than cache memory. However, it has significantly higher access latency.

- Stores running applications, operating system data, and background processes.
- When data is not available in L1, L2, or L3 cache, the processor accesses DRAM.
- Frequent DRAM accesses cause delays, power consumption, and reduced application responsiveness.

Redesign: Use faster and higher-bandwidth mobile DRAM, such as an improved LPDDR memory generation, along with better memory controllers, prefetching, and memory scheduling.

Proposed Hierarchical Memory Redesign

The redesigned memory hierarchy can be organized as:

CPU Core → L1 Cache → L2 Cache → Shared L3 Cache → High-Speed DRAM → Storage

Proposed Improvements

1. **L1 Cache:**
Maintain a small and ultra-fast L1 cache with separate instruction and data caches for immediate processor requirements.
2. **L2 Cache:**
Increase the size of private L2 caches to retain more recently used application data and reduce L3 accesses.
3. **L3 Cache:**
Use a larger shared L3 cache with intelligent allocation for active applications during app switching.
4. **DRAM:**
Use high-bandwidth, low-latency DRAM and an optimized memory controller to reduce memory access delays.
5. **Prefetching:**
Predict and load data likely to be required next, especially during application switching.
6. **Cache Replacement Policy:**
Prioritize frequently and recently used application data to prevent useful data from being removed unnecessarily.
7. **Application-Aware Memory Management:**
The operating system can identify foreground and recently used applications and retain their important data in cache whenever possible.

2. Cloud Server Memory Bottleneck

Introduction

A cloud service provider is experiencing high latency in database queries. One major reason can be a memory bottleneck, where the processor spends significant time waiting for data. Two important mechanisms involved in memory performance are the cache hierarchy and virtual memory paging. While cache hierarchy improves speed by keeping frequently used data close to the processor, virtual memory paging manages limited physical memory by moving data between RAM and storage.

A suitable combination of high-speed cache, large DRAM, and fast storage can reduce database query latency and improve server throughput.

Cache Hierarchy

A cache hierarchy consists of multiple levels of memory with different speeds and capacities:

CPU → L1 Cache → L2 Cache → L3 Cache → DRAM

L1 Cache

- Smallest and fastest cache.
- Stores immediately required instructions and data.
- Provides very low access latency.

L2 Cache

- Larger than L1 but slightly slower.
- Stores frequently reused database data and instructions.
- Reduces the number of accesses to slower memory.

L3 Cache

- Larger and usually shared among multiple processor cores.
- Useful in multi-core cloud servers where several database queries execute simultaneously.
- Improves data sharing and reduces DRAM access.

Advantages of Cache Hierarchy

- Very low latency for frequently accessed data.
- Reduces processor waiting time.
- Improves database query execution speed.
- Increases overall instruction throughput.

Limitation

Cache capacity is limited. Large databases cannot be completely stored in cache, so the system still depends on DRAM and storage.

Virtual Memory Paging

Virtual memory allows each process to use more memory than the available physical RAM by dividing memory into fixed-size **pages**.

When a required page is not present in physical RAM, a **page fault** occurs. The operating system must load the page from storage into RAM.

Memory flow during a page fault:

CPU Request → Page Not in RAM → Page Fault → Storage Access → RAM → CPU

Advantages of Virtual Memory

- Supports large database workloads.
- Allows multiple applications and virtual machines to run simultaneously.
- Provides memory isolation and protection.
- Makes efficient use of available physical memory.

Disadvantages

- Page faults create high latency.
- Frequent paging can cause **thrashing**.
- Database queries become slower when pages are repeatedly moved between RAM and storage.
- Traditional storage devices are much slower than cache and DRAM.

Comparison: Cache Hierarchy vs Virtual Memory Paging

Feature	Cache Hierarchy	Virtual Memory Paging
Main Purpose	Reduce memory access latency	Extend and manage available memory
Memory Levels	L1, L2, L3 cache	DRAM and secondary storage
Speed	Very fast	Slower during page faults
Capacity	Small to medium	Very large virtual address space

Feature	Cache Hierarchy	Virtual Memory Paging
Managed By	Hardware	Operating system and hardware
Impact on Database Queries	Improves frequently accessed data access	Can increase latency if excessive paging occurs
Main Problem	Limited capacity	High page-fault latency

Recommended Memory Improvements

1. Increase Server DRAM Capacity

The cloud server should use a larger amount of high-speed **DDR5 or other modern server-class DRAM**. More physical memory allows a larger portion of the database, indexes, and frequently accessed pages to remain in RAM.

Benefit: Reduces page faults and improves query response time.

2. Use Large L3 Cache and Modern Multi-Core Processors

A processor with a larger shared L3 cache can store frequently accessed database data closer to the CPU.

Benefit: Reduces repeated DRAM accesses and improves performance for concurrent database queries.

3. Use NVMe SSDs for Paging and Storage

If paging cannot be completely avoided, the paging system should use **high-speed NVMe SSD storage** instead of slower traditional hard drives.

Benefit: Reduces page-fault recovery time and improves database performance.

4. Use Memory-Aware Database Caching

The database should maintain frequently accessed data, indexes, and query results in memory.

Benefit: Reduces repeated disk access and improves cache hit rates.

5. Use Memory Compression and Efficient Page Management

The operating system can compress less frequently used memory pages before moving them to storage. Efficient page replacement algorithms can also reduce unnecessary page faults.

Benefit: Improves memory utilization and reduces paging overhead.

6. Consider High-Speed Persistent Memory Technologies

For workloads requiring large memory capacity with faster access than traditional storage, the provider can evaluate **persistent-memory or memory-tiering technologies** where appropriate.

Benefit: Provides an additional memory tier between DRAM and storage for large datasets.

3. Autonomous Vehicle Data Processing

Introduction

An autonomous vehicle must continuously process large amounts of real-time data from cameras, LiDAR, radar, ultrasonic sensors, GPS, and other sensors. This data is used for object detection, lane detection, path planning, and vehicle control. Since safety-critical decisions must be made within very short time intervals, the memory system must provide high bandwidth, low latency, and high reliability.

A suitable memory hierarchy can be designed by evaluating SRAM, DRAM, and MRAM according to their speed, capacity, power consumption, and non-volatility.

1. SRAM (Static Random-Access Memory)

SRAM is very fast memory commonly used for processor caches and small high-speed buffers.

Advantages

- Very low access latency.
- High read and write speed.
- Suitable for real-time and frequently accessed data.
- Does not require periodic refreshing.

Disadvantages

- High cost per bit.
- Lower storage capacity compared to DRAM.
- Consumes more chip area.

Role in Autonomous Vehicles

SRAM should store:

- Immediate sensor-processing data.
- Frequently used AI model parameters.

- CPU/GPU/AI accelerator cache data.
- Critical control instructions.
- Temporary high-priority buffers.

2. DRAM (Dynamic Random-Access Memory)

DRAM provides much larger capacity and is used as the main working memory.

Advantages

- High storage capacity.
- Lower cost per bit than SRAM.
- Supports high bandwidth.
- Suitable for storing large sensor datasets.

Disadvantages

- Higher latency than SRAM.
- Requires periodic refreshing.
- Volatile, so data is lost when power is removed.

Role in Autonomous Vehicles

DRAM should store:

- Camera frames.
- LiDAR point-cloud data.
- Radar information.
- Large AI models.
- Intermediate neural-network results.
- Maps and navigation data.

3. MRAM (Magnetoresistive Random-Access Memory)

MRAM is a non-volatile memory that retains data even when power is removed.

Advantages

- Non-volatile memory.
- Fast access compared with conventional storage.

- High endurance.
- Low standby power consumption.
- Useful for storing critical information safely.

Disadvantages

- More expensive than DRAM.
- Generally has lower capacity than main DRAM for large datasets.
- May not provide the same large-scale bandwidth as dedicated high-performance DRAM.

Role in Autonomous Vehicles

MRAM can store:

- Safety-critical system data.
- Vehicle configuration information.
- Important control parameters.
- Frequently required firmware.
- Emergency logs and sensor-event records.

Conclusion: MRAM is highly suitable for reliable, non-volatile storage of critical data.

Comparison of SRAM, DRAM, and MRAM

Feature	SRAM	DRAM	MRAM
Access Speed	Very High	High	High
Latency	Very Low	Moderate	Low
Capacity	Low	Very High	Medium
Cost per Bit	High	Low	High
Volatility	Volatile	Volatile	Non-Volatile
Power Requirement	High chip area usage	Requires refresh	Low standby power
Best Use	Cache and critical buffers	Main sensor-data memory	Persistent critical data

Proposed Memory Hierarchy

The following memory hierarchy is recommended:

Sensors → High-Speed Input Buffers → SRAM → High-Bandwidth DRAM → MRAM → Long-Term Storage

Level 1: SRAM – Lowest Latency

Small SRAM buffers and caches should be placed close to the CPU, GPU, and AI accelerators.

Used for:

- Immediate sensor data.
- Frequently accessed instructions.
- Real-time control variables.
- AI processing buffers.

Benefit: Minimizes latency for safety-critical decisions.

Level 2: High-Bandwidth DRAM – Main Working Memory

A large DRAM memory should act as the primary workspace for sensor fusion and AI processing.

Used for:

- Multiple camera streams.
- LiDAR point clouds.
- Radar data.
- Intermediate AI computations.
- High-resolution maps.

Benefit: Provides the high capacity and bandwidth needed to process multiple sensors simultaneously.

Level 3: MRAM – Persistent Fast Memory

MRAM should provide a reliable non-volatile layer for critical information.

Used for:

- Emergency vehicle-state data.
- Safety parameters.
- System configuration.

- Important logs.
- Firmware and recovery information.

Benefit: Retains critical data during power failure or unexpected system shutdown.

4. AI Inference Edge Device

Introduction

An edge AI device performs AI inference locally instead of sending all data to a cloud server. Examples include smart cameras, wearable devices, drones, and IoT systems. Such devices require fast model loading, low inference latency, and low power consumption.

To design an efficient memory system, NAND Flash, DRAM, and RRAM (Resistive RAM) can be compared based on speed, capacity, power consumption, and data persistence.

1. NAND Flash

NAND Flash is a non-volatile memory commonly used for permanent storage.

Advantages

- High storage capacity.
- Low cost per bit.
- Non-volatile, so AI models remain stored when power is turned off.
- Low standby power consumption.

Disadvantages

- Slower than DRAM and RRAM.
- Higher latency during model loading.
- Limited write endurance compared with some emerging memory technologies.

Role in Edge AI

NAND Flash is suitable for storing:

- Trained AI models.
- Operating system files.
- Application programs.

- Configuration data.
- Large datasets and model updates.

Conclusion: NAND Flash is best used as the large-capacity storage layer.

2. DRAM

DRAM is volatile, high-speed main memory used while the AI model is actively running.

Advantages

- Faster than NAND Flash.
- High capacity.
- High bandwidth for AI computations.
- Suitable for storing model weights and intermediate inference results.

Disadvantages

- Volatile, so data is lost when power is removed.
- Requires continuous refreshing, which increases power consumption.
- Model data must be reloaded after restart.

Role in Edge AI

DRAM is suitable for:

- Active AI model weights.
- Intermediate neural-network activations.
- Input data buffers.
- Temporary computation results.

Conclusion: DRAM is the main working memory for AI inference but consumes more power than non-volatile alternatives.

3. RRAM

RRAM, or Resistive Random-Access Memory, is an emerging non-volatile memory technology that stores data by changing the resistance of memory cells.

Advantages

- Non-volatile.
- Low standby power.

- Fast access compared with NAND Flash.
- Can provide lower energy consumption for frequently accessed stored data.
- Suitable for compact edge devices.

Disadvantages

- Generally more expensive and less widely available than conventional NAND Flash and DRAM.
- Capacity and manufacturing maturity may be limited.
- Technology characteristics can vary depending on implementation.

Role in Edge AI

RRAM can be used for:

- Frequently used AI model weights.
- Frequently accessed parameters.
- Fast model loading.
- Persistent low-power storage.

Conclusion: RRAM can act as an intermediate memory layer between NAND Flash and DRAM.

5. High-Performance Gaming Console: L3 Cache vs DRAM Throughput and Memory Hierarchy Enhancements

Introduction

A high-performance gaming console may experience **frame drops during large scene loading** because the CPU and GPU must rapidly access textures, geometry, game logic, animation data, shaders, and other assets. If required data is not available in the faster cache hierarchy, the processor must access DRAM or even storage, increasing latency and reducing the rate at which frames can be prepared.

To solve this problem, the performance of **L3 cache and DRAM** must be analyzed, followed by improvements to the overall memory hierarchy.

1. Role of L3 Cache

The **L3 cache** is a large, high-speed cache located between the CPU cores and main memory. It is generally shared among multiple CPU cores.

Advantages of L3 Cache

- Lower latency than DRAM.
- Stores frequently accessed game data.
- Reduces repeated access to main memory.
- Supports data sharing among CPU cores.
- Improves CPU performance during physics, AI, animation, and scene management.

During gaming, L3 cache can hold frequently reused data such as:

- Player and object states.
- Physics calculations.
- AI data.
- Animation information.
- Frequently used instructions.
- Scene metadata.

Limitation

L3 cache has limited capacity. Large textures, detailed 3D models, and complete game scenes may be too large to fit completely in L3 cache. Therefore, cache misses can still force access to DRAM.

2. Role of DRAM

DRAM is the main working memory of the gaming console and provides much larger capacity than L3 cache.

Advantages of DRAM

- Large capacity for game assets.
- High bandwidth for CPU and GPU workloads.
- Stores large textures, models, audio, and scene data.
- Supports multiple game processes simultaneously.

Limitations

- Higher access latency than L3 cache.

- Can become a bottleneck when many CPU and GPU requests occur simultaneously.
- Insufficient bandwidth can cause delays while loading large scenes.

During large scene loading, DRAM may need to supply data for:

- High-resolution textures.
- 3D geometry.
- Shader resources.
- Animation data.
- Audio resources.
- World and map information.

If the memory controller cannot supply this data quickly enough, the rendering pipeline may wait, resulting in **stuttering or frame drops**.

3. Comparison of L3 Cache and DRAM

Feature	L3 Cache	DRAM
Location	Close to CPU cores	Main system memory
Latency	Low	Higher
Capacity	Moderate	Very High
Throughput	High for cached data	High overall bandwidth
Best Use	Frequently reused data	Large game assets
Cost per Bit	High	Lower
Main Limitation	Limited capacity	Higher latency

Analysis

L3 cache is more effective when the same data is accessed repeatedly because it provides faster access and avoids DRAM delays. DRAM is essential for storing large amounts of game data but can cause performance problems when demand exceeds available bandwidth.

Therefore, the goal is to **maximize cache hit rates while increasing DRAM bandwidth and reducing unnecessary memory transfers**.

MARKS ALLOCATION

	Understanding of Industry Problem (20%)	Application of Engineering Concepts (25%)	Solution Design (25%)	Use of Tools (15%)	Analysis (10%)
Q1					
Q2					
Q3					
Q4					
Q5					

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools,	Appropriate use of tools with correct	Limited or inconsistent use of tools	Minimal or incorrect use of tools

		technologies, and real data	implementati on		
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis